FIELDCOMMAND

Incident Management System v1.0

COMPLETE USER MANUAL

v1.0

ICS/NIMS All-Hazards · Amateur Radio EMCOMM · Public Safety

August 07, 2026

Table of Contents

1 Introduction & System Overview

When a disaster takes down commercial infrastructure — cellular networks, internet, power — the tools modern emergency management depends on disappear precisely when they are needed most. Incident logs revert to paper. Resource tracking becomes a whiteboard. Incident Command System (ICS) forms are filled out by hand, photocopied, and hand-carried between rooms. Situational awareness degrades to whatever one person can hold in their head. Every organization that has worked a major activation knows this failure mode, and most have simply accepted it as the cost of doing business.

FieldCommand IMS was built to eliminate that failure mode. It is a complete, self-contained incident management platform that carries its own network, its own server, and its own tools — and it operates with no internet connection, no cellular service, and no outside infrastructure of any kind. The cell towers are down, the internet is gone, and you are running on a generator in a parking lot. That is exactly when you need incident management software most — and that is exactly when FieldCommand IMS is designed to perform.

1.1 What FieldCommand IMS Is

FieldCommand IMS is a **complete ICS/NIMS all-hazards incident management system** — not simply an amateur radio or Emergency Communications (EMCOMM) tool. It manages the full lifecycle of any incident from initial response through demobilization using standard ICS forms and workflows. It runs on a Raspberry Pi 5 server and broadcasts its own private Wi-Fi access point. Any smartphone, tablet, or laptop that joins that network immediately has access to the full suite of 48 tools through a standard web browser — no app installation, no accounts, and no per-device configuration required.

When internet connectivity is available, live features activate automatically: National Weather Service (NWS) weather alerts, APRS-IS, animated Next Generation Radar (NEXRAD) radar, and High Frequency (HF) propagation data. If connectivity is lost at any point, all core tools continue without interruption. The system degrades gracefully and recovers automatically.

1.2 How FieldCommand IMS Differs From Other ICS Platforms

Platforms such as WebEOC, E-Team, NIMSIAP, National Incident Management System (NIMS) Logic, and E-iSuite deliver powerful incident management capability — but every one of them requires a working internet connection and functioning server infrastructure. They are cloud-dependent by design. When a major disaster disables that infrastructure, they fail with it — at exactly the moment capability is most critical.

FieldCommand IMS is built on the opposite assumption: infrastructure will fail, and incident management capability must survive that failure. It carries its own infrastructure — server, network, storage, and tools — in a single deployable package. It requires no IT department, carries no licensing fees, and does not fail when the internet fails.

The feature that sets FieldCommand IMS apart from every platform in this category is its native integration of amateur radio and public safety communications directly into the incident management workflow. No other ICS platform includes built-in net control logging, Federal Communications Commission (FCC) callsign validation against the full national licensee database, Automatic Packet Reporting System (APRS) tactical mapping, Winlink radio email, JS8Call HF messaging, or Amateur Packet Radio Network (AMPRNet) gateway capability. These are not add-ons — they are core features fully integrated with the ICS platform so that radio traffic, net logs, and check-in data flow directly into ICS-309 communications logs, ICS-214 activity logs, and the Incident Action Plan (IAP).

PLATFORM	OFFLINE?	SELF-CONTAINED?	NATIVE EMCOMM?	COST
FieldCommand IMS	✓ Fully offline	✓ Own network + server	✓ Full EMCOMM	Free / open source
WebEOC	✗ Cloud only	✗ Needs infrastructure	✗ None	$10,000–50,000/yr
E-Team / Veoci	✗ Cloud only	✗ Needs infrastructure	✗ None	Subscription
NIMSIAP	✗ Internet required	✗ Requires connectivity	✗ None	Free
E-iSuite	■ Limited	■ Laptop, no network	✗ None	License fee

1.3 System Architecture

COMPONENT	FUNCTION	DEFAULT IP
FieldCommand Pi 5 16GB	Primary server · all 48 web tools · 12+ background services · RAID-1 NVMe storage · EMCOMM-NET Wi-Fi AP	192.168.50.1
44Net Gateway Pi 5 (optional)	AMPRNet WireGuard tunnel · callsign-authenticated access · Part 97 access log · isolated from primary server	192.168.50.2
Wi-Fi router (primary)	DHCP · AiMesh controller · dual Wide Area Network (WAN) management · EMCOMM-NET Service Set Identifier (SSID) broadcast. Recommended: ASUS RT-BE58 Go.	192.168.50.254
AiMesh nodes (optional)	EMCOMM-NET coverage extension · seamless roaming. Recommended: additional ASUS RT-BE58 Go units.	DHCP assigned
UniFi Switch Lite 16 Power over Ethernet (PoE) (recommended)	Wired backbone · powers PoE devices · connects all Pi units, router, cellular antenna, and workstations	DHCP assigned
Operator workstations	Any device with a modern browser: smartphones, tablets, laptops, Raspberry Pi 500 desktops, Windows or macOS laptops	DHCP assigned

■ The default server address is 192.168.50.1. This is configurable during installation. All documentation uses this default — substitute your address wherever 192.168.50.1 appears if your deployment differs.

1.4 Capability Summary

CAPABILITY AREA	TOOLS PROVIDED
Incident Management	Five-section ICS structure. Complete IAP form set: ICS-202 through ICS-221. Live T-card resource board. Pre-planned event templates for 6 incident types. Incident archive, restore, and scenario/exercise mode.
ICS Forms & IAP	Full ICS form set with digital signature capture on all prepared-by and approved-by fields. One-click IAP Portable Document Format (PDF) compilation with title page and section dividers. Print center for on-site IAP packages.
Federal Emergency Management Agency (FEMA) Documentation	FEMA Public Assistance (PA) cost tracking: Force Account Labor with fringe, Equipment with 2025 FEMA rate schedule (45 built-in rates), Materials, and Contracts. Real-time cost dashboard. ICS-214 import. Project Worksheet text export.

CAPABILITY AREA	TOOLS PROVIDED
Amateur Radio EMCOMM	Net control logger with FCC callsign auto-fill from offline 800,000+ licensee database. APRS tactical map. Winlink and Pat radio email. JS8Call HF digital. AMPRNet 44Net gateway. National Traffic System (NTS) radiogram generator.
Public Safety Comms	Trunked/P25 net logger with radio ID check-in. EMA member ID lookup. ICS-309 export. Observer mode for read-only net monitoring on any device.
Situational Awareness	Animated NEXRAD radar (WAN required). Live NWS weather alerts. GPS-tracked resource map with status-coded SVG markers. Offline APRS tactical map. SARTopo GeoJSON import. HF propagation tool.
Personnel & Check-In	Member roster with certifications, radio IDs, and Quick Response (QR) code check-in codes. QR/barcode camera scan check-in using native BarcodeDetector API. ICS-211 check-in log. Digital ID card generator.
WAN & Connectivity	Dual WAN source support: any cellular modem, hotspot, or satellite dish. Preferred/fallback role configuration. Configurable detection per source. Automatic failover. Real-time WAN status with 30-second polling.
Reference & Offline Content	Kiwix offline Wikipedia and reference library. Radio frequency cheat sheets. Hospital and facilities directory. Repeater database. Channel library. NIMS resource typing library. ICS position checklists.

2 Getting Started — Connecting to FieldCommand

http://192.168.50.1

Every tool in FieldCommand IMS is a web page served from the Pi. No app installation is required on any device. Any modern browser works — Chrome, Firefox, Safari, Edge, or the built-in browser on any smartphone or tablet.

2.1 First Connection

1. Power on the FieldCommand Pi and wait approximately 45 seconds for all services to start. The system is ready when the EMCOMM-NET Wi-Fi network appears in the device scan list.
2. On any smartphone, tablet, or laptop — open Wi-Fi settings and connect to **EMCOMM-NET**. The password is on the equipment case label and in the Installation Guide. No other credentials are required.
3. Open a browser and navigate to **http://192.168.50.1**. The FieldCommand dashboard loads immediately.
4. Bookmark the dashboard Uniform Resource Locator (URL). On a smartphone, tap **Share** → **Add to Home Screen** to create an icon — it opens like a native app.
5. If the dashboard does not load, confirm you are connected to EMCOMM-NET and not to another Wi-Fi network. The Pi does not relay traffic to the internet — if your device switches to a different network automatically, disable auto-join on other networks during activations.

2.2 Dashboard Modes

The dashboard has three selectable modes that filter the tool card grid. Switching modes does not affect any data — all tools remain accessible in every mode.

MODE	TOOLS SHOWN PROMINENTLY	BEST FOR
Amateur Radio	Net control logger, Federal Communications Commission (FCC) callsign lookup, Automatic Packet Reporting System (APRS) map, Winlink, JS8Call, Amateur Packet Radio Network (AMPRNet) gateway, High Frequency (HF) propagation, National Traffic System (NTS) radiogram	Amateur radio net operations, ARES/RACES activations
Public Safety	public-safety net logger, radio ID roster, resource map, Wide Area Network (WAN) status, weather radar, hospital directory, facilities	Public safety radio nets, public-service check-in operations
ICS	Full ICS platform, Incident Action Plan (IAP) forms, T-card resource board, personnel check-in, Federal Emergency Management Agency (FEMA) cost documentation, event templates, cost dashboard	Any active incident requiring ICS structure

2.3 WAN Status Indicators

Each mode includes a live WAN status card (not a top-of-page bar). It polls every 30 seconds and changes color automatically.

INDICATOR	MEANING	CORE TOOLS AFFECTED?
■ Green — Cellular	Primary internet source active	None — fully operational
■ Blue — Satellite	Satellite failover active	None — fully operational

INDICATOR	MEANING	CORE TOOLS AFFECTED?
■ Red — Offline	No internet — all core tools continue	None — fully operational
■ AMPRNet UP	44Net WireGuard tunnel active	N/A
■ AMPRNet DOWN	44Net tunnel down or gateway Pi unreachable	N/A

A red WAN indicator means internet-dependent features are paused — National Weather Service (NWS) radar, APRS-IS, HF propagation. It does NOT mean FieldCommand IMS is unavailable. All ICS tools, forms, net loggers, roster, and local maps run normally offline.

3 Organization Setup & Station Configuration

http://192.168.50.1/setup.html

Before using FieldCommand IMS for the first time, complete the Organization Setup to configure your station identity, Service Set Identifier (SSID), and default settings. These values appear in Incident Command System (ICS) form headers, net logs, and printed documents.

3.1 Setup Fields

FIELD	WHAT IT CONFIGURES
Organization name	Full name of your agency or group — appears in ICS form headers and cover pages
Short name / abbreviation	Used in compact displays, footer lines, and net log headers
Primary callsign	Club or personal Federal Communications Commission (FCC) callsign — used for Amateur Packet Radio Network (AMPRNet) authentication and net logs
Station grid square	Maidenhead grid locator (4 or 6 character) — used for High Frequency (HF) propagation calculations and Automatic Packet Reporting System (APRS) position reports
Default incident name	Pre-fills the incident name field when creating new incidents
Wi-Fi network name (SSID)	Default is EMCOMM-NET — change if you operate multiple FieldCommand systems simultaneously and need to distinguish them
Server address	Default 192.168.50.1 — change only if this conflicts with your existing network
Time zone	Used for all timestamps in logs and forms — set to your deployment time zone

3.2 Offline vs. Online Feature Table

FEATURE	OFFLINE ?	NOTES WHEN OFFLINE
ICS platform — all five sections	✓ Full	No change
All ICS forms and IAP	✓ Full	No change
T-card resource board	✓ Full	No change
FEMA cost tracking	✓ Full	No change
Event templates	✓ Full	No change
IAP PDF compilation	✓ Full	No change
Digital signatures on forms	✓ Full	No change
Net control loggers (both)	✓ Full	No change
FCC callsign lookup	✓ Full	Uses local SQLite database — no internet needed
Member roster and QR check-in	✓ Full	No change
GPS resource map	✓ Full	No change

FEATURE	OFFLINE ?	NOTES WHEN OFFLINE
Barcode / QR scan check-in	✓ Full	No change
Offline tactical APRS map	✓ Full	Map tiles local — RF APRS still works
Kiwix reference library	✓ Full	No change
NWS weather alerts	■ WAN	Paused — last alert cached and displayed
NEXRAD animated radar	■ WAN	Tiles unavailable — offline banner shown
APRS-IS internet feed	■ WAN	RF APRS continues — internet feed pauses
HF propagation data	■ WAN	Last retrieved data shown until WAN returns
AMPRNet / 44Net tunnel	■ WAN	Tunnel drops — local network unaffected
Winlink Telnet sessions	■ WAN	RF Winlink (VARA/Pactor) continues

4 Member Roster & Quick Response (QR) code Check-In Codes http://192.168.50.1/roster.html

The Member Roster is the central personnel database. It stores every member, mutual-aid visitor, and regular participant with their identifiers, certifications, equipment capabilities, and a unique barcode check-in code that enables rapid QR-scan check-in at any activation.

4.1 Roster Fields

FIELD	PURPOSE
Member ID	Your organization's internal identifier (e.g. MEM-042). Primary key.
Barcode ID	QR/barcode check-in code — defaults to Member ID on first import. Can be overridden with a facility badge number or any custom code.
Callsign	Federal Communications Commission (FCC) amateur radio callsign. Leave blank for non-licensed members.
Radio ID	Trunked/P25 radio unit number if applicable.
Name	First and last name.
Role	Primary function: Operator, Net Control, Logistics, Medical, etc.
Member type	Member · Visitor · Mutual Aid — affects check-in form behavior.
Certifications	ICS-100/200/300/400/700/800, EmComm I/II, CPR, First Aid, CERT.
Equipment	High Frequency (HF), Very High Frequency (VHF), digital modes, Automatic Packet Reporting System (APRS), Winlink, go-box, generator, vehicle.
License class	Technician, General, Amateur Extra — displayed on net log entries.
Phone / Email	Contact information — stored locally, never transmitted externally.
Grid square	Maidenhead grid locator for RF planning.

4.2 Adding and Editing Members

1. Click **+ Add Member** at the top of the roster page.
2. Enter at minimum a Member ID and name. Add callsign, radio ID, or both if applicable.
3. Check all applicable certifications and equipment capabilities.
4. Click **Save**. The member appears immediately in the roster.
5. To edit an existing member, click their card to open the detail view, make changes, and click **Save**.
6. To delete a member, open their detail view and click **Delete**. This removes them from the roster but does not delete historical net log entries.

4.3 Importing Members via CSV

Import your existing member list using ■ **Import CSV**. The CSV must have a **member_id** column. All other columns are optional. Existing member IDs are updated; new IDs are added. Existing members not in the import file are left unchanged.

Required: **member_id**

Optional: **callsign** · **radio_id** · **first_name** · **last_name** · **role** · **phone** · **email** · **grid** · **license_class**

4.4 QR Check-In Codes

Every member has a personal QR code that enables instant check-in at any activation using the Scan Check-In page (see Chapter 10). The QR button on each roster card opens a modal with that member's personal code.

1. Click the **QR** button on any member card.
2. The modal shows the QR code. When internet is available, a scannable QR image is generated (via the Google Charts image API — note this Google service is deprecated and may not always render). When offline, the member ID is displayed in large text as a manual-entry fallback.
3. Click ■ **Print** to open a clean print-ready page with the QR, the member's name, and the FieldCommand check-in Uniform Resource Locator (URL).
4. Click ■ **Save PNG** to download the QR code image — members can save it to their phone's home screen or print it on their ID badge.

> ■ The barcode_id field defaults to the member_id on first import. If your organization uses physical facility badge scanners, set barcode_id to match the badge number — the scan check-in will then auto-fill from a badge swipe.

5 Incident Management — Creating and Managing Incidents http://192.168.50.1/incident.html

All Incident Command System (ICS) work in FieldCommand IMS is organized around an **incident**. An incident holds the ICS forms, resource T-cards, personnel check-ins, cost tracking, and net log associations for one activation or exercise. You can have multiple incidents in the system simultaneously — only one is designated the **active incident** at any time.

5.1 Creating an Incident

1. Navigate to **Incident Management** from the dashboard.
2. Click **+ New Incident**.
3. Enter the incident name, incident number (if assigned by your agency), incident type, and initial operational period.
4. Select the incident commander from your roster, or type a name if the Incident Commander (IC) is not in the roster. For incidents operating under **Unified Command**, see Section 5.2 below.
5. Click **Create Incident**. The incident opens immediately and becomes the active incident. The incident name appears in the dashboard header and in all form headers.

5.2 Command Structure — Single Incident Commander vs. Unified Command

ICS supports two command structures, and FieldCommand IMS accommodates both without any configuration change. The difference is a workflow convention, not a software setting.

STRUCTURE	WHEN USED	HOW TO CONFIGURE IN FIELDCOMMAND
Single IC	One agency or jurisdiction has clear authority. Most activations involving a single responding organization.	Enter the IC's name or callsign in the Incident Commander field when creating the incident. The ICS-203 org chart will show one IC at the top of the command structure.
Unified Command	Two or more agencies or jurisdictions share command authority — common in multi-agency responses, complex disasters, or incidents crossing jurisdictional boundaries. Each contributing agency provides an IC-equivalent to the UC group. All agencies still work from a single Incident Action Plan (IAP).	Enter "Unified Command" or the UC group designation in the Incident Commander field. List each UC member in the ICS-203 Organization Assignment List under the Unified Command section. All ICS forms, T-cards, and cost documentation continue to function identically — the UC structure is a labeling and documentation convention within the same incident record.

Under Unified Command, each participating agency still develops its own objectives, but those objectives are reconciled into a single IAP. The Planning Section Chief role in the ICS-203 is typically filled by one agency on a rotating basis or by consensus. FieldCommand IMS supports multiple ICS-204 Assignment Lists — one per branch or division — which accommodates multi-agency operational structures naturally.

5.3 Incident Types

When creating an incident, pick the type that best matches the activation from a large categorized dropdown (about 32 options grouped into seven categories). The type labels the incident badge throughout the interface. All types use the same underlying ICS form set — the type is an organizational label, not a functional restriction.

CATEGORY	EXAMPLE TYPES
Natural Hazards	Winter Storm, Flooding, Tornado / Severe Weather, Earthquake, Wildfire, Heat Emergency, Drought
Technological	Hazmat / Chemical Spill, Transportation Accident, Structure Fire, Power Outage / Infrastructure, Dam Failure, Nuclear / Radiological
Human-Caused	Mass Casualty Incident, Active Threat, Civil Disturbance, Terrorism
Search & Rescue	Wilderness, Urban, Water, Missing Person — Dementia / Memory, Missing Person — Child
Public Health	Disease Outbreak / Pandemic, Mass Casualty — Medical, Public Health Emergency
Planned Events	Planned Event — Public Safety, Planned Event — EMCOMM Exercise, Drill / Training Exercise
Other	Mutual Aid Request, EOC Activation, Other / All-Hazards

Any incident can be flagged as a drill/exercise. Beta Reset (under incident settings) wipes exercise data while preserving the roster, channel library, and repeater database.

5.4 Operational Periods

ICS organizes work into **operational periods** — typically 12 or 24-hour blocks. FieldCommand IMS tracks the current operational period and uses it as the default for new T-cards, check-ins, and form entries. To advance to a new operational period, click **+ New Period** in the incident header. Prior period data is retained and viewable.

5.5 Archive, Restore, and Delete

After an incident closes, the full data package can be archived to a Universal Serial Bus (USB) drive for long-term storage and removed from the Pi's active storage.

ACTION	WHAT IT DOES	REVERSIBLE?
Archive to USB	Writes a complete JSON package (all forms, T-cards, check-ins, costs, net log associations) to /media/fieldcommand/backup/incidents/ on the labelled USB drive. Marks the incident as archived on the Pi.	✓ Yes — restore any time
Restore from USB	Reads the JSON archive from the USB drive and re-inserts all data. Restores the incident to fully active status.	✓ Yes
Hard delete from Pi	Permanently removes the incident and all associated data from the Pi SSD. Run only after confirming the archive is on USB.	✗ Permanent

■ The USB drive must be labelled FIELDCOMMAND (all caps). Any USB drive with this label is recognized automatically. A LaCie Rugged or similar ruggedized 1TB USB-C drive is recommended for field use.

5.6 Exercise / Scenario Mode

Exercises and training scenarios can be tagged as ■ **Scenario** when the incident is created. Scenario incidents display a yellow badge throughout the interface so operators always know they are in training mode. A **Beta Reset** wipes all scenario data — incidents, forms, costs, check-ins, T-cards, and meetings — while preserving the roster, hospital directory, channel library, and repeater database. This returns the system to a clean state for the next exercise without losing any permanent configuration data.

6 Pre-Planned Event Templates http://192.168.50.1/event_templates.html

Event Templates allow you to pre-configure a complete incident setup — including objectives, default resource types, communications channels, and Incident Command System (ICS) organization — and activate it in seconds at the start of an activation. Templates eliminate the repetitive setup work for recurring incident types and ensure consistency across activations.

6.1 Built-In Templates

FieldCommand IMS ships with six built-in templates covering the most common incident types. Each can be used as-is or edited to match your local protocols.

TEMPLATE	PRE-CONFIGURED CONTENTS
Shelter Activation	Shelter management objectives, cot and supply resource types, registration and medical channels, Red Cross coordination section
Search & Rescue	Search and Rescue (SAR) objectives, field team and K9 resource types, search sector channels, base camp and medical branches
Severe Weather	Damage assessment objectives, utility and debris resource types, shelter and Emergency Operations Center (EOC) coordination channels, public information branch
Mass Gathering / Event	Crowd management objectives, medical and security resource types, venue and dispatch channels, medical and operations branches
HazMat / Spill	Decon and zoning objectives, HazMat team resource types, hot/warm/cold zone channels, safety officer emphasis
Planned Exercise / Drill	Training objectives, evaluator and observer resource types, exercise control channel, scenario-tagged (■)

6.2 Activating a Template

1. Navigate to **Event Templates** from the dashboard.
2. Click the **Activate** tab. Browse the template gallery.
3. Click the template you want. A configuration panel opens.
4. Enter the incident name, date/time, and any other details you want to customize before activating.
5. Click **Activate Template**. FieldCommand IMS creates a new incident pre-loaded with all objectives, resource types, channels, and ICS org structure from the template, then shows success links to open the new incident (it does not navigate there automatically).

6.3 Creating and Editing Templates

1. Click the **Manage Templates** tab.
2. To edit a template, click its card. The edit modal opens with sections for **Objectives**, **Resources**, **Channels**, and **Organization**.
3. Edit any field live. Objectives and resource types are listed — click **+ Add** to add items, click the × to remove them.
4. Click **Save Template** when finished.
5. To export a template as a JSON file for sharing with other FieldCommand systems, click **Export JSON**. To import a JSON template, click **Import JSON** and select the file.

6. Built-in templates can be edited but not deleted. Custom templates can be deleted with the ■ **Delete** button.

7 Amateur Radio Net Control Logger http://192.168.50.1/netcontrol.html

The Amateur Net Control Logger provides complete net management for licensed amateur radio operations: ARES, RACES, AUXCOMM, and any other amateur net. It replaces paper net logs with a live digital record that feeds directly into the ICS-309 communications log.

7.1 Opening a Net

1. Navigate to **Net Control Logger** from the dashboard.
2. Enter the net name, frequency, and mode (SSB / FM / Digital / Other).
3. Verify the date and time are correct — these become the net open timestamp.
4. Click **Create Net**. The net becomes active and the live elapsed timer starts.
5. Share the ■ **Observer Link** with section chiefs or served agencies so they can monitor the net in read-only view on their own devices. The observer page auto-refreshes every 15 seconds.

7.2 Logging Check-Ins

When a station checks in, type their callsign in the callsign field. FieldCommand IMS looks up the callsign instantly in the **offline Federal Communications Commission (FCC) database** — 800,000+ amateur licensees stored locally on the Pi — and fills the operator's name and license class automatically. If the station is on your roster, their member ID is filled as well. No internet required.

1. Type the station's callsign. The name fills automatically once you finish entering it.
2. Enter location or remarks if desired.
3. Click **LOG ENTRY** or press Enter. The entry appears with a timestamp.
4. To check a station out, click **Check Out** on their row. Checkout time and participation duration are recorded automatically, rounded up to the nearest quarter-hour for reimbursement documentation.
5. For traffic, use the traffic entry at the bottom — enter precedence, addressee, and a brief description.

7.3 Closing a Net and Exporting ICS-309

1. Click **Close Net**. All remaining checked-in stations are automatically checked out with the net close timestamp.
2. Net open time, close time, and total duration appear in the header.
3. Click ■ **Export ICS-309** to download the complete ICS-309 Communications Log as a formatted document ready for the Incident Action Plan (IAP).
4. The ICS-309 includes: net name, frequency, mode, open/close times, total duration, and a full check-in table with individual participation durations rounded up to the nearest quarter-hour.

> ■ The Dead Man's Switch (deadmans.html) monitors net activity and sounds an alert if no check-in is logged within a configurable time window. It is designed for safety monitoring during Search and Rescue (SAR) and field operations where radio silence may indicate an emergency. See Chapter 12.

8 Public Safety Net Logger http://192.168.50.1/starcom.html

The Public Safety Net Logger handles trunked radio, P25, and other public safety radio systems where check-in is by radio ID rather than callsign. It is designed for interoperability exercises, served-agency support, and any net where participants may not hold amateur licenses.

FEATURE	AMATEUR NET LOGGER	PUBLIC SAFETY NET LOGGER
Primary ID	Federal Communications Commission (FCC) callsign — auto-filled from local database	Radio ID (unit number)
License check	FCC ULS lookup confirms active license	None required
Roster lookup	By callsign → name and member ID	By radio ID → name and member ID
ICS-309 format	Callsign in station column	Radio ID in station column
Typical use	ARES / RACES / AUXCOMM	Public safety agency nets, interop exercises

Operation follows the same workflow as the Amateur Net Logger — open the net, log check-ins by radio ID, and export ICS-309 at close. Observer Mode is also available: click ■ **Observer Link** to share a read-only, auto-refreshing view of the active net with any device on EMCOMM-NET.

> ■ Participants who hold both an amateur license and a public safety radio ID should check into the public safety net by radio ID only. If they need to participate in a concurrent amateur net on the same incident, they check into the Amateur Net Logger separately by callsign. This maintains the integrity of both logs.

9 Observer Mode — Read-Only Net View http://192.168.50.1/observer.html

Observer Mode provides a read-only, auto-refreshing view of any active net. It is designed for section chiefs, served agency liaisons, Emergency Operations Center (EOC) duty officers, or anyone who needs to monitor net activity without access to the Net Control workstation.

9.1 Accessing Observer Mode

1. The Net Control operator clicks ■ **Observer Link** on the active net page.
2. The Uniform Resource Locator (URL) is copied to the clipboard. Share it via Winlink, JS8Call, in person, or announce it over the net.
3. Any device on EMCOMM-NET opens the link in a browser. No login required.
4. The observer page shows the net name, the current check-in list with callsigns/names/times, and the traffic log. (The observer header does not currently repeat the frequency/mode or a live elapsed timer.) It refreshes automatically every 15 seconds.

9.2 What Observers Can and Cannot Do

OBSERVERS CAN	OBSERVERS CANNOT
View all check-ins in real time	Add or remove check-ins
View net open time and elapsed duration	Close the net
View the traffic log	Add traffic entries
Refresh the page to get latest data	Change any net settings
Bookmark the observer URL for re-use	Access the Net Control view

10 Barcode & Quick Response (QR) code Scan Check-In

http://192.168.50.1/scan_checkin.html

The Scan Check-In page enables rapid personnel check-in at activations using a smartphone or tablet camera. It uses the browser's native BarcodeDetector API — no external library, no CDN dependency, works fully offline. It supports QR codes, Code 128, Code 39, EAN-13, Data Matrix, PDF417, and Aztec formats.

10.1 Two Check-In Methods

METHOD	HOW IT WORKS	BEST FOR
Camera scan	Point the device camera at the member's QR code or barcode. Detection is automatic at 4 frames per second. The same code is debounced for 3 seconds to prevent duplicate submissions.	High-throughput check-in stations; members with printed QR codes
Manual ID entry	Type a member ID, callsign, or radio ID in the manual entry field and press Enter or click Look Up. Same roster lookup as camera scan.	When camera unavailable; walk-in personnel without a QR code

10.2 Check-In Workflow

1. Navigate to **Scan Check-In** from the dashboard.
2. Click ■ **Start Camera**. If multiple cameras are present (front/rear), select the rear-facing camera from the dropdown.
3. Point the camera at the member's QR code. The system detects it automatically and looks up the member in the roster.
4. If the member is found: their name, agency, and suggested Incident Command System (ICS) position auto-fill with green borders. Review the fields — edit if needed.
5. If the member is not in the roster: a "not found" banner appears. Their scanned ID fills the ID field. Fill in name and agency manually.
6. Click ✓ **Check In**. A full-screen success confirmation appears with the member's name and check-in time. The device vibrates briefly on phones that support it.
7. Click **Scan Next Person** to clear the form and check in the next member. The session history panel at the bottom shows the last 10 check-ins.

10.3 Roster Lookup Order

When a code is scanned or a manual ID is entered, the system searches the roster in this order until a match is found:

SEARCH ORDER	FIELD SEARCHED	EXAMPLE VALUE
1st	barcode_id	MEM-042 (or badge number if overridden)
2nd	member_id	MEM-042
3rd	callsign	KE4CON
4th	radio_id	412

> BarcodeDetector is supported on Chrome (Android and desktop), Edge, and Samsung Internet. It is not supported on iOS Safari or Firefox. On unsupported browsers, the camera scan button is disabled and the manual entry fallback works normally.

11 Federal Communications Commission (FCC) Callsign Lookup http://192.168.50.1/callsign.html

The FCC Callsign Lookup tool provides instant offline access to the full national FCC amateur radio licensee database — over 800,000 active licensees. The database is stored locally in SQLite on the Pi and is used automatically by the Net Control Logger, Scan Check-In, and Incident Action Plan (IAP) forms. No internet connection is required.

11.1 Manual Lookup

1. Navigate to **Callsign Lookup** from the dashboard.
2. Type a callsign and press **Enter** or click the lookup button — results are not live-as-you-type.
3. The result shows: callsign, name, license class, license status, expiration date, and mailing address city/state.
4. Click **Copy** to copy the callsign to the clipboard, or **Add to Roster** to create a roster entry from the FCC record.

11.2 Automatic Lookup in Other Tools

The callsign database is queried automatically in three other places:

WHERE	HOW IT WORKS
Net Control Logger	When you enter a callsign in the check-in field, the name and license class fill automatically. A red border appears if the callsign is not found or the license is expired.
Scan Check-In	After a Quick Response (QR) code scan or manual entry, if the code is a callsign, the FCC record is used to fill the name field (falling back to the roster).
ICS-213	The "From Callsign" field on the general message form fills the operator name from the FCC database.

■ The FCC database is refreshed weekly. To update, download the FCC ULS database export from wireless.fcc.gov/uls and run the database import script as described in the Installation Guide.

12 Dead Man's Switch http://192.168.50.1/deadmans.html

The Dead Man's Switch monitors net activity and triggers an audible alert if no check-in is logged within a configurable time window. It is designed for field operations — particularly search and rescue — where radio silence beyond a set interval may indicate a field team emergency.

12.1 Configuration

SETTING	DESCRIPTION
Check-in interval	Maximum time allowed between net check-ins before the alert fires. Common values: 10, 15, 30 minutes. Configurable per operation.
Alert type	Audible tone through the browser (requires speaker). Visual flashing banner on the Dead Man's page.
Reset	Any new check-in in the Net Control Logger resets the timer automatically.

12.2 Operating the Dead Man's Switch

1. Open the Dead Man's Switch page on a dedicated monitor or tablet at the Net Control position or Emergency Operations Center (EOC) duty station.
2. Select the active net from the dropdown.
3. Set the check-in interval and click **Start Monitoring**.
4. The countdown timer displays time remaining until alert.
5. If the timer reaches zero, the audible alert fires and the page flashes red. Acknowledge with **Acknowledge** and investigate.
6. Any check-in logged in the Net Control Logger resets the timer automatically.
7. Click **Stop Monitoring** when the net closes.

13 Tactical Automatic Packet Reporting System (APRS) Map

http://192.168.50.1/tactical.html

The Tactical Map is a Leaflet-based map showing APRS station positions from both RF APRS (always available) and APRS-IS internet feed (when Wide Area Network (WAN) is up). It is the primary situational awareness tool for tracking field teams, vehicles, and mobile resources.

> Offline caveat: the map is fully offline only when the Leaflet library and map tiles are served locally from the Pi. In the current build, Leaflet and the default base tiles load from the internet (a Content Delivery Network), so without a WAN the base map may not draw — station markers and overlays still plot on whatever base is available. Vendor Leaflet and pre-download tiles locally for a true no-internet map.

13.1 Map Layers

LAYER	SOURCE	REQUIRES WAN?
Map tiles	OpenStreetMap / CartoDB Dark	Cached locally for offline use
APRS-IS stations	Internet APRS-IS feed	✓ WAN required
RF APRS stations	Local Terminal Node Controller (TNC) or direwolf connected to Pi	✗ Offline — always available
SARTopo overlays	GeoJSON import from sartopo_import.html	✗ Offline once imported
Weather alert polygons	National Weather Service (NWS) API	✓ WAN required

13.2 Station Markers

Each APRS station is displayed with its standard APRS symbol. Clicking a marker opens a popup showing: callsign, timestamp of last beacon, speed and course (if moving), altitude, and the raw APRS comment field. Stations are color-coded by age — bright for recent beacons, faded for stations not heard in more than 30 minutes.

13.3 SARTopo GeoJSON Import

Planning data from SARTopo (search sectors, assignments, exclusion zones) can be exported as GeoJSON and imported into the tactical map using the SARTopo Import page (sartopo_import.html). Once imported, the overlay persists on the map through browser refreshes.

1. In SARTopo, export your map or layer as GeoJSON.
2. Navigate to **SARTopo Import** from the reference section of the dashboard.
3. Click **Choose File** and select the GeoJSON file.
4. Click **Import**. The overlay appears on the tactical map immediately.
5. To remove the overlay, return to SARTopo Import and click **Clear Overlay**.

14 GPS-Tracked Resource Map http://192.168.50.1/resource_map.html

The Resource Map displays the current Global Positioning System (GPS) positions of all Incident Command System (ICS) resources for the active incident. Each resource is drawn as a color-coded SVG pin whose color reflects its current status. Resources without a GPS position appear as dashed circles in the sidebar so the Operations Section can see what still needs to be positioned.

14.1 Map Controls

CONTROL	FUNCTION
Incident selector	Choose which incident's resources to display. The last selected incident is remembered between sessions.
■ Refresh	Reload all resource positions from the server.
■ My Location	Place a marker at your current device GPS position and zoom to it. Useful for confirming map accuracy against your known position.
Auto-refresh 30s	Enable automatic position refresh every 30 seconds. Use when resources are actively moving and updating their positions.

14.2 Marker Colors by Status

COLOR	STATUS
■ Green	Available
■ Blue	Assigned
■ Light blue	Staging
■ Red	Out of Service or En Route
Dashed circle	No GPS position set yet

14.3 Setting a Resource Position

Click any resource in the sidebar — or click an existing pin on the map — to open the Set Position modal. Three methods are available:

METHOD	HOW TO USE	BEST FOR
Device GPS	Click ■ **Use Device GPS**. The browser requests your device's GPS position. Accuracy (±meters) is shown.	Field operator at the resource's location using a phone
Click map to place	Click ■ **Click Map to Place**. The modal closes and the cursor changes to a crosshair. Click anywhere on the map to set the position.	Incident Commander (IC) or Planning placing resources on a map by known location
Manual coordinates	Type decimal latitude and longitude directly into the fields.	Position known from a paper map, GPS receiver, or radio report

Optionally enter a **Location Label** (e.g. "Division Alpha staging area") to display a human-readable description below the coordinates. Click **Save Position**. The pin appears on the map immediately. Click **Clear GPS** to remove the position.

15 Incident Command System (ICS) Platform Overview — Five-Section Structure http://192.168.50.1/incident.html

FieldCommand IMS implements the full five-section ICS structure defined by National Incident Management System (NIMS). Each section has its own page with the forms, tools, and resource displays appropriate to that function. All sections work from the same underlying incident database — a resource added in Operations appears immediately in Logistics and Finance.

ICS SECTION	PAGE	KEY TOOLS
Command	incident.html	Incident overview, general info, ICS-201 initial briefing, ICS-202 objectives, ICS-207 org chart, ICS-208 safety message, position checklists, meeting scheduler. Note: the Safety Officer (SOFR) and Public Information Officer (PIO) are Command Staff — they report directly to the Incident Commander (IC), not to a section.
Operations	ics/operations.html	T-card resource board by type and status, assignment tracking, Global Positioning System (GPS) resource map, ICS-204 assignment lists. Note: ICS-204 is developed by the Resources Unit (Planning) but distributed to and used by Operations supervisors.
Planning	iap.html	Incident Action Plan (IAP) assembly, ICS-202 objectives, ICS-203 org assignment list, ICS-209 status summary, ICS-211 check-in list, ICS-221 demobilization, Planning P cycle. Note: Planning assembles and distributes the IAP but does not develop all forms in it — see the form table in Chapter 17 for development responsibility.
Logistics	ics-form.html	ICS-205 radio comms plan (Communications Unit Leader (COML)), ICS-205A comms list (COML), ICS-206 medical plan (Medical Unit Leader (MEDL)), ICS-218 support vehicle inventory (GSUL), ICS-309 communications log (COML), facilities directory, channel library. Note: the Communications Unit and Medical Unit are in the Logistics Service Branch. They develop ICS-205, 205A, 206, and 309 — these forms are then included in the IAP by the Planning Section.
Finance / Admin	fema_costs.html	Federal Emergency Management Agency (FEMA) Public Assistance (PA) cost tracking, Force Account Labor with fringe, Equipment with FEMA rate schedule, Materials, Contracts, cost dashboard, ICS-214 activity log import. Note: ICS-214 is completed by all supervisors in every section — Finance/Admin collects them for cost documentation and records.

15.1 Command Structure — Single IC or Unified Command

The ICS-203 Organization Assignment List is where the command structure is formally documented. For a Single IC incident, the IC's name appears at the top of the org chart. For a Unified Command incident, the Unified Command (UC) member agencies are listed collectively at the Command level. FieldCommand IMS handles both structures identically — the ICS-203, ICS-202, and all downstream forms work the same way regardless of command structure. See Chapter 5.2 for guidance on configuring each structure.

15.2 General Info — ICS-201 Initial Briefing

The General Info page (general_info.html) holds the ICS-201 Initial Incident Briefing data: situation summary, initial response actions, current organization, and resource summary. It is typically completed by the first-arriving IC and updated through the first operational period.

15.3 Preflight Deployment Checklist

The Preflight page (preflight.html) provides a comprehensive GO/CAUTION/NO-GO checklist for verifying system and data readiness before every activation. It combines automated server checks that run without any operator input with a manual section-by-section checklist covering hardware, power, communications, personnel, logistics, safety, and agency coordination. A rolling verdict at the top of the page updates in real time as items are checked.

Data Readiness — Automated Checks

The first section of the preflight checklist — Data Readiness — runs automatically when the page loads and whenever the operator clicks Re-Check. It queries the FieldCommand server and verifies six critical data items:

CHECK ITEM	WHAT IS VERIFIED	REQUIRED FOR GO?
Organization Setup	Org name and callsign have been entered in the setup page.	Yes — required
FCC Callsign Database	FCC amateur license database is present and less than 14 days old. Shows record count and age in days.	Yes — required
Member Roster	At least one member has been imported or entered in the roster.	Recommended
Repeater Database	Repeater data has been loaded from a RepeaterBook CSV or manual entry. Shows total count and how many have map coordinates.	Recommended
Channel Library	At least one channel has been added to the channel library.	Recommended
Active Incident	An incident has been created and set to active status.	Recommended

■ Each auto-check item shows live server detail inline — for example "✓ 847,223 records, 3.2 days old" or "✗ 0 repeaters loaded." Items that fail auto-check are automatically set to NO-GO. Operators can override any item manually if circumstances warrant. Each NO-GO item description includes the page URL to navigate to and fix it.

Manual Checklist Sections

SECTION	COVERS
■ Power Systems	Shore/generator power, battery backup/UPS, solar, fuel, power distribution
■ Communications Equipment	HF radio, VHF/UHF radio, go-kit radios, antennas, repeater access, Winlink, JS8Call
■ Computing & Software	Server boot, FCC database, nginx, API server, health monitor, Graywolf APRS, Kiwix, Wi-Fi AP
■ Personnel & Staffing	Net Control operator, alternate NCS, agency liaison, operator briefing, roster check-in
■ Logistics & Supplies	Food and water, first aid, weather protection, spare parts, lighting, paper backup forms
■ Safety & Security	Lightning protection, RF exposure, CO detector, emergency comms plan, contact list

SECTION	COVERS
■ Agency Coordination	EOC check-in, frequencies confirmed, ICP location, resource request process, demobilization plan

How to Use the Preflight Checklist

1. Open **http://192.168.50.1/preflight.html** on any EMCOMM-NET device. The Data Readiness section runs automatically — review any NO-GO items first.
2. For each NO-GO data item, click the link in the item description to navigate to the fix page (roster.html, repeaters.html, channel_library.html, or incident.html). Return to preflight and click ■ **Auto-Check** to re-verify.
3. Work through each manual section. For each item, click ✓ **GO**, ■ **CAUTION**, or ✕ **NO-GO**. Add a note in the text field if needed.
4. The verdict banner updates in real time. **GO** (green) means all required items are confirmed. **CAUTION** (amber) means optional items need attention. **NO-GO** (red) means a required item has failed — do not activate.
5. Click **Save State** to store the current checklist to browser storage for the shift record. Click **Export Report** to download a JSON summary for documentation.

16 Incident Command System (ICS) Operations Section — T-Card Resource Board

http://192.168.50.1/ics/operations.html

The T-Card Resource Board is the Operations Section's primary resource tracking tool. It mirrors the physical ICS T-card system used in traditional incident management but adds live status, assignment tracking, Global Positioning System (GPS) integration, and direct export to ICS-204 assignment lists.

16.1 Resource Types

Resources are organized by National Incident Management System (NIMS) type. The Resource Types library (resource_types.html) defines the types available in your system. FieldCommand ships with standard NIMS types pre-loaded: Crew, Engine, Helicopter, Dozer, Water Tender, Medical Unit, Strike Team, Task Force, and Single Resource. Custom types can be added for your specific operations.

16.2 T-Card Fields

FIELD	DESCRIPTION
Resource name	Unit designation (Engine 12, Team Alpha, Unit 412)
Resource type	NIMS type from the resource types library
Status	Available · Assigned · Staging · Out of Service · En Route
Assignment	Current task or division/group assignment
Leader	Supervisor name or callsign
Personnel count	Number of personnel on this resource
Category	Agency, mutual aid source, or contractor
Daily cost / Rate	Used by Federal Emergency Management Agency (FEMA) cost tracking and cost dashboard
GPS position	Latitude, longitude, location label — set from Resource Map
Operational period	Which OP period this T-card was created for

16.3 Personnel Assigned to a Resource

On a physical ICS T-card, the back of the card lists the individual personnel assigned to that resource — not just the count, but the names, ICS positions, and contact information for each person on the crew, unit, or team. FieldCommand IMS replicates this with a dedicated Personnel tab on every T-card detail panel. This roster is linked to the master Member Roster so that names auto-fill from existing records.

FIELD	DESCRIPTION
Name	Full name of the assigned person. Begin typing to see roster autocomplete suggestions — selecting a suggestion fills name, callsign, and agency automatically.
ICS Position on Resource	The specific role this person fills on this resource. Examples: Crew Boss, Paramedic/EMT, Emergency Radio Operator, Driver/Operator, Equipment Operator. Different from their overall ICS position on the org chart.

FIELD	DESCRIPTION
Agency	Home agency or mutual aid source. Auto-filled from roster when name is matched.
Contact / Callsign	Radio callsign, channel, or phone number for this person. Auto-filled from roster when a licensed amateur is matched.

16.4 Using the Personnel Tab

1. Click any T-card on the resource board to open the detail panel.
2. Click the ■ **PERSONNEL** tab at the top of the panel. The count in parentheses shows how many personnel are currently listed.
3. Begin typing a name in the Name field. If the person is in the roster, their name appears as a blue suggestion chip below the field. Click the chip to auto-fill name, callsign, and agency.
4. Select the person's ICS position on this specific resource from the dropdown. This is their functional role on the resource — Crew Boss, EMT, Operator, etc.
5. Fill in agency and contact if not auto-filled from the roster.
6. Click ✓ **Add to Resource**. The person appears in the personnel list and the personnel count on the T-card board updates automatically.
7. To remove a person, click the ✕ button on their row. The count updates automatically.

> ■ The Personnel tab roster is separate from the check-in list (ICS-211). The personnel list answers "who is on this resource?" — the check-in list answers "who has checked into this incident?" A person can be in both, or either. For example, a mutual-aid crew might all be listed on their engine's T-card personnel tab but only the crew boss may check into the ICS-211 on behalf of the crew.

16.5 Adding and Managing Resources

1. On the T-Card board, click **+ Add Resource**.
2. Fill in the resource name, type, status, and assignment.
3. Click **Save**. The T-card appears on the board.
4. To change status, click the status badge on the T-card and select the new status.
5. To reassign, click the assignment field and type the new assignment.
6. To set or update GPS position, click ■ **Set Position** or go to the Resource Map (Chapter 14).
7. To close out a resource, set status to **Out of Service** and fill in the demobilization notes.

17 Incident Command System (ICS) Planning Section & Incident Action Plan (IAP) Assembly http://192.168.50.1/iap.html

The Planning Section manages the Incident Action Plan — the written work plan for each operational period. FieldCommand IMS covers the full ICS form set needed to build a complete IAP, with digital signature capture and one-click Portable Document Format (PDF) compilation.

17.1 IAP Form Set

The table below shows every ICS form in FieldCommand IMS with two key attributes: who is responsible for **developing** it, and which section it is **distributed through**. These are not always the same. Several forms are developed by Logistics unit leaders but assembled into the IAP and distributed by the Planning Section — a distinction that confuses many operators and is explained in the notes below the table.

FORM	TITLE	DEVELOPED BY	DISTRIBUTED THROUGH / IN IAP?
ICS-202	Incident Objectives	Planning Section Chief / IC	Planning · ✓ IAP
ICS-203	Organization Assignment List	Resources Unit Leader (RESL) — Planning	Planning · ✓ IAP
ICS-204	Assignment List	Resources Unit Leader (RESL) — Planning	Operations (from Planning) · ✓ IAP
ICS-205	Radio Communications Plan	Comms Unit Leader (Communications Unit Leader (COML)) — **Logistics**/Service Branch	Via Planning into IAP · ✓ IAP
ICS-205A	Communications List	Comms Unit Leader (COML) — **Logistics**/Service Branch	Supplemental — often in IAP
ICS-206	Medical Plan	Medical Unit Leader (MEDL) — **Logistics**/Service Branch	Via Planning into IAP · ✓ IAP
ICS-207	Incident Organization Chart	Resources Unit Leader (RESL) — Planning	Command display · ✓ IAP
ICS-208	Safety Message / Plan	Safety Officer (SOFR) — Command Staff	Via Planning into IAP · ✓ IAP
ICS-209	Incident Status Summary	Situation Unit Leader (SITL) — Planning	Sent to MAC/EOC — not in IAP
ICS-211	Incident Check-In List	Resources Unit Leader (RESL) — Planning	Entry point recorders — not in IAP
ICS-213	General Message	Any ICS position	Any section — not in IAP
ICS-214	Activity Log	All supervisors — every section	Documentation only — not in IAP
ICS-215A	IAP Safety Analysis	Safety Officer (SOFR) — Command Staff, with Ops input	Via Planning into IAP · ✓ IAP

FORM	TITLE	DEVELOPED BY	DISTRIBUTED THROUGH / IN IAP?
ICS-218	Support Vehicle/Equipment Inventory	Ground Support Unit Leader (GSUL) — Logistics/Support Branch	Logistics internal — not in IAP
ICS-221	Demobilization Check-Out	Demob Unit Leader (DMOB) — Planning	Signed by demob resources — not in IAP
ICS-309	Communications Log	Comms Unit Leader (COML) / any operator — **Logistics**	Comms unit records — not in IAP

17.1a ICS Doctrine Notes — Forms That Cross Section Lines

Three forms are commonly misattributed in the field because they appear in the IAP (which Planning assembles) but are developed by Logistics unit leaders. Understanding the distinction matters for knowing who to ask when a form needs to be corrected or updated.

FORM	THE DOCTRINE DISTINCTION
ICS-205 Radio Comms Plan	The **Communications Unit Leader (COML)** develops the ICS-205. The COML reports to the Service Branch Director under the Logistics Section. The completed ICS-205 is handed to the Planning Section, which includes it in the IAP and distributes it at the operational period briefing. If the ICS-205 needs to be corrected, go to the COML — not the Planning Section Chief.
ICS-205A Comms List	Also developed by the **COML** under Logistics. The 205A is a supplemental contact directory — not always formally in the IAP but typically attached. Same attribution rule: developed in Logistics, Via Planning.
ICS-206 Medical Plan	The **Medical Unit Leader (MEDL)** develops the ICS-206. The MEDL also reports to the Service Branch Director under Logistics. The completed ICS-206 goes to Planning for IAP inclusion. The Safety Officer (SOFR) must review and concur with the ICS-206 before it is finalized — their signature appears on the form.

17.2 Digital Signature Capture

All 16 Prepared By and Approved By fields across the ICS form set support digital signature capture. A signature pad appears below each signature field on every form. Signatures are stored as white-background PNG images and are embedded in the printed form and the IAP PDF.

1. Click on any **Prepared By** or **Approved By** signature field.
2. A signature pad opens below the field.
3. Sign using mouse, touchscreen, or stylus. The signature supports pressure sensitivity on compatible styluses.
4. Click **Accept**. A preview of the signature appears below the field.
5. The signature is saved automatically and restored if the page is reloaded.
6. To clear a signature, click **Clear** on the signature pad.

17.3 IAP Assembly Page — Print and Save

The IAP Assembly page (iap.html) lets the Planning Section select which ICS forms to include in the operational period package, then either print the IAP immediately or save it as a portable file for off-site printing. Two export options are available:

OPTION	WHEN TO USE	HOW IT WORKS
■ Print IAP	A printer is available on EMCOMM-NET or attached to the operator device.	Opens a formatted print window in a new browser tab with a cover page, table of contents, and links to each selected ICS form. The browser print dialog opens automatically.
■ Save as PDF (recommended)	No printer on site, or the IAP needs to be taken off-site, emailed to the EOC, or archived. Use this as the primary save option.	Posts the selected form list to the FieldCommand server, which generates a proper 8.5" × 11" PDF using ReportLab — the same engine used for all FieldCommand documents. Downloads as IAP_[Incident]_Period[N]_[Date].pdf. Opens in any PDF viewer on any device. Guaranteed correct page layout, margins, and pagination. Requires the FieldCommand Pi to be running (always true on EMCOMM-NET).
■ Save as HTML (fallback)	Server is unreachable, or the operator prefers a lightweight portable file.	Downloads a self-contained HTML file with the IAP cover page and form index directly from the browser — no server call required. Open in any browser and use File → Print (Ctrl+P / Cmd+P) to print. Note: HTML printing can produce inconsistent page layout depending on the browser and its print settings. Use PDF when layout matters.

1. Navigate to **http://192.168.50.1/iap.html** or click IAP in the Planning section.
2. Select the operational period and form variant (FEMA / USCG / NWCG).
3. Check the forms to include. Required forms are pre-checked. Typically include ICS-202, 203, 204, 205, 205A, 206, 207, and 208.
4. **To print on-site:** click ■ **Print IAP**. A formatted print window opens in a new tab. Select the site printer.
5. **To save off-site (PDF — recommended):** click ■ **Save as PDF**. The Pi generates a proper PDF and it downloads automatically as `IAP_[Incident]_Period[N]_[Date].pdf`. Copy to USB or email — opens in any PDF viewer anywhere.
6. **To save off-site (HTML — fallback):** click ■ **Save as HTML**. Downloads immediately without a server call. Open in any browser and use File → Print. Use this only if the PDF button fails.

17.4 IAP PDF Compilation

The IAP Compile page (iap_compile.html) assembles all completed ICS forms into a single server-generated PDF with embedded signatures, a cover page, and section dividers. This is the full distributable IAP package.

1. Navigate to **IAP Compile** from the dashboard or the Planning section.
2. The page shows a checklist of all ICS forms for the active incident. Completed forms show a checkmark. Incomplete or missing forms are flagged.
3. Select which forms to include. Required forms for a standard IAP: ICS-202, 203, 204, 205, 206, 207, 208.
4. Click **Download IAP PDF**. The Pi server generates the PDF — this takes 5 to 15 seconds depending on how many forms are included.
5. The PDF downloads automatically. It can be printed from any device on EMCOMM-NET or saved to USB for off-site printing.

> ■ The IAP PDF is generated on the Pi server and can be downloaded from any device on EMCOMM-NET — no printer needs to be connected to the device that generates it. The downloaded PDF is fully self-contained and can be printed at any location.

18 Federal Emergency Management Agency (FEMA) Public Assistance (PA) Cost Documentation http://192.168.50.1/fema_costs.html

The FEMA PA Cost Documentation module tracks all reimbursable costs for FEMA Public Assistance (PA) grant documentation. It covers the four FEMA cost categories: Force Account Labor, Equipment, Materials, and Contracts. All entries are tied to the active incident and operational period.

18.1 Force Account Labor

Force Account Labor tracks regular-time and overtime hours for your organization's employees and volunteers. For FEMA PA purposes, only overtime hours are reimbursable for regular employees; all hours may be reimbursable for volunteers depending on your state's policies.

FIELD	FEMA REQUIREMENT
Employee name	Full name as on payroll records
Job title / class	Official job classification
Regular hours	Regular-time hours worked on incident
Overtime hours	Overtime hours — primary FEMA reimbursable category
Hourly rate	Regular hourly rate from payroll
Fringe benefit rate	Fringe as percentage of wages (FEMA requires fringe documentation)
Total with fringe	Calculated automatically: (hours × rate) × (1 + fringe%)
ICS-214 import	Import hours directly from a completed ICS-214 Activity Log

18.2 Equipment

Equipment entries use the FEMA Schedule of Equipment Rates. FieldCommand IMS includes all 44 standard FEMA equipment categories pre-loaded from the current year schedule. Select the equipment type and the applicable rate fills automatically.

1. Click **+ Add Equipment Entry**.
2. Click ■ **Lookup** to open the rate picker. Search by equipment type — the FEMA rate fills automatically.
3. Enter hours used and the equipment identifier (unit number or VIN).
4. Add the operator name if the operator is separate from the equipment.
5. The total calculates automatically: hours × FEMA rate.

18.3 Materials and Contracts

Materials entries document consumable supplies purchased specifically for the incident. Contracts entries document work performed by contractors. Both require: vendor name, description, purchase order or contract number, unit cost, quantity, and total. Attach receipts and PO documentation as noted — FEMA requires source documentation for all cost claims.

18.4 Project Worksheet Export

Click ■ **Export Project Worksheet (PW) Text** to generate a formatted text summary of all cost entries suitable for copying into the FEMA Grants Portal or attaching to a Project Worksheet. The export includes incident name, dates, itemized costs by category, and total claimed amount.

> ■ FEMA equipment rates are updated annually. The system displays a reminder when the loaded rates are more than one year old. Update rates using the FEMA Equipment Rates page (Chapter 19).

19 Federal Emergency Management Agency (FEMA) Equipment Rate Schedule http://192.168.50.1/fema_rates.html

The FEMA Equipment Rates page maintains the Schedule of Equipment Rates used by the FEMA cost documentation module. FieldCommand IMS ships with all 44 standard FEMA equipment categories pre-loaded from the 2025 schedule. Rates can be updated annually as FEMA publishes new schedules.

19.1 Viewing and Searching Rates

1. Navigate to **FEMA Equipment Rates** from the Finance section.
2. The table shows all 44 categories with FEMA code, description, unit (hour or day), and current rate.
3. Use the search box to filter by equipment type or FEMA code.
4. A **Rate Year** badge shows which year's schedule is loaded. A yellow warning appears when rates are more than one year old.

19.2 Updating Rates

1. Download the current FEMA Schedule of Equipment Rates from fema.gov.
2. On the FEMA Equipment Rates page, click ✏ **Edit** on any row to update that rate. Enter the new rate and click **Save**.
3. After updating all changed rates, click **Update Rate Year** and enter the current year. This clears the stale-rates warning.
4. Rates are stored in the local database and used immediately by the FEMA cost documentation module.

20 Cost Tracking Dashboard http://192.168.50.1/cost_dashboard.html

The Cost Dashboard provides a real-time financial overview of the active incident. It aggregates all Federal Emergency Management Agency (FEMA) cost entries and T-card daily rates into a summary view updated automatically every two minutes.

20.1 Dashboard Panels

PANEL	WHAT IT SHOWS
Total incident cost	Running total of all documented costs across all categories. Large tile at the top — visible at a glance from across the room.
Cost by category	Horizontal breakdown bars: Labor · Equipment · Materials · Contracts. Shows both dollar amount and percentage of total for each category.
Resources by type	Personnel count and cost contribution from T-card resources, grouped by resource type.
Cost projection	Extrapolated cost at fixed horizons (12 hr, 24 hr, 72 hr, and 7 days) based on the current burn rate. Useful for budget management during multi-day incidents.
Budget tracker	Compare actual costs against an authorized budget amount. Enter the budget figure to see remaining balance and percentage used.
Per-period breakdown	Cost summary broken out by operational period — useful for after-action documentation and FEMA Project Worksheet (PW) preparation.

■ The cost dashboard draws data from both the FEMA cost entries (fema_costs.html) and the daily rate fields on T-cards. Set daily or hourly rates on each T-card to see resource costs reflected automatically in the dashboard.

21 Personnel Accountability http://192.168.50.1/accountability.html

Personnel accountability is a fundamental National Incident Management System (NIMS)/ICS safety requirement. At any moment during an incident, the Incident Commander and Safety Officer must be able to answer two questions with certainty: **Who is on this incident?** and **Where is each person right now?** Failure to maintain accountability has resulted in responder fatalities at wildland fires, structural collapses, and other incidents where conditions changed rapidly and personnel could not be located.

FieldCommand IMS implements a two-level accountability system that mirrors the Incident Command System (ICS) doctrine: incident-level accountability through the ICS-211 Check-In List, and resource-level accountability through the T-card personnel roster. The Personnel Accountability page integrates both into a single Safety Officer tool with Personnel Accountability Report (PAR) (Personnel Accountability Report) tracking, check-out recording, last-known-location logging, and automatic cross-referencing between the two lists.

21.1 The Two Levels of Personnel Accountability

LEVEL — TOOL	WHAT IT TRACKS	MAINTAINED BY
Incident level ICS-211 Check-In List (checkin.html / scan_checkin.html)	Who has formally checked into this incident? When did each person arrive? Have they checked out? Every person on the incident must have an ICS-211 entry regardless of agency, role, or resource assignment.	Check-in recorder at entry points. Resources Unit Leader (RESL) maintains the master list.
Resource level T-Card Personnel tab	Who is assigned to each specific resource? Which crew or unit is each person on? Where should each crew member be right now?	Ops Section Chief (OSC); crew supervisors update T-card lists.

Both levels are required — neither alone is sufficient. The ICS-211 tells you everyone who is on the incident. The T-card personnel roster tells you which resource each person is assigned to and where they should be. The cross-reference between the two catches gaps in both directions: people who checked in but were never assigned to a resource, and people on a T-card who never completed formal ICS-211 check-in.

21.2 ICS-211 Check-In — Incident-Level Accountability

Every person arriving at the incident — regardless of agency, role, or resource assignment — must complete ICS-211 check-in. This is the foundation of personnel accountability. There are three ways to check in:

METHOD	BEST FOR
Camera scan (scan_checkin.html)	High-throughput check-in stations and members with personal Quick Response (QR) codes or barcodes. Scan detects automatically at 4 frames per second — no typing required.
Manual form entry (checkin.html)	Walk-in personnel, mutual aid, or any person without a Quick Response (QR) code. Full ICS-211 form with ICS position dropdown and agency field.
Scan page manual fallback (scan_checkin.html)	When camera is unavailable — type the member ID, callsign, or radio ID directly into the manual entry field on the scan page.

21.3 ICS-211 Check-Out

Check-out is as important as check-in. When a person leaves the incident — for any reason — they must be checked out so the accountability system reflects the actual population on the incident. Unchecked-out personnel create false positives in PAR counts and can trigger unnecessary searches.

1. Navigate to **Personnel Accountability** from the dashboard, or to **Incident Check-In** (checkin.html).
2. Find the person in the ICS-211 check-in list.
3. Click **Check Out** on their row. The checkout timestamp is recorded automatically. They are removed from the active accountability count.
4. Their record is retained for documentation — they do not disappear from the list, they move to the "Checked Out" section.
5. For mass checkout at incident close, use **Conduct PAR** → **Check All Out** from the Personnel Accountability page.

21.4 T-Card Personnel Roster — Resource-Level Accountability

Every resource T-card should have a complete list of the personnel assigned to it. This is maintained by the Operations Section and crew supervisors, and it gives the Safety Officer resource-level accountability: not just "Jones is on this incident" but "Jones is assigned to Engine 12, which is assigned to Division Alpha."

1. Navigate to **ICS Operations** from the ICS section.
2. Click any T-card to open its detail panel.
3. Click the ■ **PERSONNEL** tab.
4. Add each person assigned to that resource — name, their position on the resource (Crew Boss, EMT, Operator, etc.), agency, and contact/callsign.
5. If the person is in the roster, type their name and click the suggestion chip — callsign and agency fill automatically.
6. Repeat for every person on every resource. The T-card board shows the personnel count on each card.
7. Update the list if personnel rotate on or off a resource during the incident.

> ■ The T-card personnel list and the ICS-211 check-in list are separate records. Adding someone to a T-card does NOT automatically check them into the ICS-211. Both must be maintained. The Cross-Reference view on the Personnel Accountability page flags anyone who is in one list but not the other.

21.5 Personnel Accountability Report (PAR)

A Personnel Accountability Report (PAR) is a formal roll call — a deliberate, time-stamped check that confirms every person on the incident is accounted for. PARs should be conducted at regular intervals (typically every operational period or whenever conditions change), at all incident transitions (shift changes, operational period changes), whenever a hazard escalates, and immediately whenever any person cannot be located.

The Personnel Accountability page (accountability.html) is the Safety Officer's primary PAR tool. It shows both the ICS-211 list and the T-card personnel roster, with PAR confirmation buttons for each person.

1. Navigate to **Personnel Accountability** from the dashboard.
2. Select the incident and operational period.
3. Click ■ **Conduct PAR**. The timestamp is recorded. The display switches to the ICS-211 list with all personnel sorted so unconfirmed personnel appear at the top.
4. Contact each supervisor or crew boss by radio and have them confirm all personnel on their resource. As each person is confirmed, click ✓ **PAR** on their row. The count updates in real time.
5. For T-card-level PAR, switch to the **T-CARD PERSONNEL** tab and confirm by resource — each resource group shows its own PAR count.

6. The ■ **UNACCOUNTED** tab shows only the personnel who have not been confirmed. If anyone cannot be accounted for after exhausting radio contact, escalate immediately to the Incident Commander (IC).
7. When all personnel are confirmed, the summary tiles show 0 unaccounted and the PAR badge updates to green.
8. To begin a new PAR cycle, click ■ **Reset PAR**. All confirmation flags are cleared and the cycle begins again.

21.6 Last Known Location

The last known location field provides an additional layer of situational awareness during a PAR. When a supervisor confirms their crew, they can also report the crew's current location — division, sector, grid coordinate, or landmark. This information displays on both the accountability page and the resource map.

To update last known location: click the ■ **Location** button on any person's row. A text field appears — enter the location (e.g. "Division Alpha, east perimeter" or "Base Camp medical tent") and click Save. The location is recorded with a timestamp.

21.7 Cross-Reference — Closing the Accountability Gap

The Cross-Reference tab compares the ICS-211 check-in list against the T-card personnel rosters and flags two specific gaps:

CONDITION	WHAT IT MEANS	ACTION REQUIRED
Checked in — not on any T-card	The person completed ICS-211 check-in but does not appear on any resource T-card. They are on the incident but not assigned to any resource. They may be floating staff, a late assignment, or the T-card may not have been updated.	Assign them to a resource and add them to the T-card, or confirm they are intentionally unassigned (e.g. Safety Officer, PIO) and note their location.
On T-card — not checked in via ICS-211	The person is listed on a resource T-card but has no ICS-211 entry. This is a **safety gap**: they are on the incident, possibly deployed to a hazardous area, but are not in the formal accountability system.	Direct them to complete ICS-211 check-in immediately. If they cannot be located, escalate to the IC.

■ Run the cross-reference at the start of each operational period and any time new personnel arrive at the incident. A clean cross-reference — no flags in either direction — means your ICS-211 and T-card rosters are in sync and every person on the incident is in both the formal check-in system and assigned to a resource.

21.8 Accountability During Rapid Deterioration

When incident conditions change rapidly — a structure becoming unstable, a fire run, a weather change, a hazmat release — the Safety Officer may need to conduct an emergency PAR immediately. The following procedure is recommended:

1. Click ■ **Conduct PAR** immediately. The timestamp establishes the start of the accountability check.
2. Transmit on the incident command channel: "*All supervisors, this is Safety, emergency PAR — report personnel count by division/group immediately.*"
3. As each supervisor reports, click ✓ **PAR** for each person on their resource in the T-CARD PERSONNEL tab.
4. Check the ■ **UNACCOUNTED** tab after each supervisor reports. If anyone remains unaccounted after all supervisors have reported, transmit their name and last known assignment immediately.

5. If a person cannot be located within five minutes, treat it as a missing responder emergency — notify the IC, halt operations in the affected area, and initiate search.

6. The Dead Man's Switch (deadmans.html) should already be active if this is a Search and Rescue (SAR) or high-risk operation — it provides an independent alert if net check-ins stop (see Chapter 12).

22 ICS-213 General Message http://192.168.50.1/ics213.html

The ICS-213 General Message is the standard Incident Command System (ICS) form for written communications between sections, agencies, or individuals during an incident. FieldCommand IMS provides a fillable ICS-213 with digital signature capture, print output, and Winlink integration.

22.1 Completing an ICS-213

1. Navigate to **ICS-213 General Message** from the dashboard.
2. Fill in: To, From (callsign auto-fills name from Federal Communications Commission (FCC) database), Subject, and Date/Time.
3. Type the message in the message body field.
4. If a reply is required, check **Reply Requested**.
5. Click the **Prepared By** signature field and sign with mouse, touch, or stylus (see Chapter 17.2 for signature instructions).
6. Click **Print** to print the completed form, or **Send via Winlink** to route the message through Winlink if a Winlink session is active.

22.2 Winlink Integration

A completed ICS-213 can be exported directly to Winlink Express or Pat as a formatted Winlink ICS-213 message. The recipient address, subject, and message body are pre-filled. See Chapter 29 for Winlink configuration.

23 ICS-214 Activity Log & ICS-309 Communications Log

http://192.168.50.1/ics214.html

The ICS-214 Activity Log records the activities and significant events for each resource or section during an operational period. Completed ICS-214 entries can be exported directly to Federal Emergency Management Agency (FEMA) Public Assistance (PA) labor cost documentation, bridging the gap between activity logging and reimbursement paperwork.

23.1 ICS-214 Activity Log

1. Navigate to **ICS-214** from the Planning section.
2. Select the resource or section this log covers.
3. For each significant activity, click **+ Add Entry**: enter the time, activity description, and resources involved.
4. At the end of the operational period, click **Prepared By** and sign the form.
5. Click **Export to FEMA Labor** to push hours from this log directly into the FEMA Force Account Labor documentation for the active incident.

23.2 ICS-309 Communications Log

There are two ways to produce an ICS-309. The **automatic** one lives on the Net Logger (netcontrol.html / starcom.html): its **Export ICS-309** button builds a Communications Log from that net — the message traffic first (from/to/time/precedence/message), then the station check-in list, with the net name, frequency/mode, and open/close times in the header. It both downloads a saved file and opens a print view. Separately, **ics309.html** is a manual-entry ICS-309 for traffic not captured in a Net Logger.

1. For a net you logged: open that net in the Net Logger and click **Export ICS-309**.
2. For manual entry: open **ics309.html** and type each log line by hand.
3. Either way, the exported ICS-309 downloads as a file (and can be printed) for the Incident Action Plan (IAP).

24 Wide Area Network (WAN) Settings & Dual-Source Internet Configuration http://192.168.50.1/wan_settings.html

FieldCommand IMS supports two simultaneous internet sources — primary and fallback — with automatic failover between them. The WAN Settings page configures how each source is detected and how the system monitors them. No carrier names or hardware brands are hardcoded — the system works with any cellular modem, phone hotspot, satellite dish, or fixed broadband.

24.1 WAN Source Configuration

FIELD	OPTIONS AND DESCRIPTION
Role	**Preferred** — the primary WAN source. Used first when available. **Fallback** — used only when the preferred source is unavailable.
Label	Human-readable name shown on the dashboard status bar: "Cellular", "Satellite", "Hotspot", etc.
Type	Cellular · Satellite · Hotspot · Fixed broadband · Other
Detection method	How the monitor determines if this source is up. See the table below for all methods.
Enabled	Toggle a source on or off without deleting its configuration.

24.2 Detection Methods

METHOD	HOW IT WORKS	BEST FOR
internet_only	Attempts an outbound internet connection. Succeeds if any internet Any internet path succeeds.	Phone hotspots, Universal Serial Bus (USB) cellular dongles, any source without a fixed gateway IP
ping	Pings a specific IP address — typically the modem or router gateway. Enter the gateway IP address.	Sources with a known gateway IP (e.g. a cellular router on 192.168.1.1)
admin_reachable	Checks if the modem's admin web interface is reachable at a given Uniform Resource Locator (URL). Faster than a full internet check for modems with local admin pages.	Cellular modems with a local admin interface

24.3 Swapping Preferred and Fallback

To swap which source is preferred and which is fallback, click the ■ **Swap Roles** button on the WAN Settings page. This takes effect immediately without restarting any services. Use this when you want to manually prefer satellite over cellular, for example, without changing your physical connections.

> The WAN monitor runs as a background service and updates the dashboard status bar every 30 seconds. If you change WAN settings, allow up to 60 seconds for the new configuration to take effect.

25 National Weather Service (NWS) Animated Radar

http://192.168.50.1/radar.html

The Next Generation Radar (NEXRAD) Radar page displays animated radar loops sourced from RadrView, with an Iowa Environmental Mesonet (IEM) NEXRAD Web Map Service (WMS) fallback if RadrView is unavailable. It requires an active internet connection to fetch new frames. The radar page is accessible from all three dashboard modes.

25.1 Radar Controls

CONTROL	FUNCTION
■ / ■ Play/Pause	Start or stop the radar animation loop
■ / ■ Step	Advance one frame at a time — useful for analyzing storm movement
Timeline scrubber	Jump to any point in the loaded radar loop
Speed selector	Slow · Medium · Fast — controls animation playback speed
Palette	Default · Dark · NOAA
Station selector	Center the map on a chosen NEXRAD station’s coverage area
Auto-refresh	Loads new radar data every 5 minutes automatically
Reflectivity legend	Color scale overlay showing dBZ values and associated precip rates

25.2 Fallback When Offline

When the WAN connection is unavailable, the radar page displays an "Offline — radar unavailable" banner and shows the last successfully loaded frame with its timestamp. This allows you to see the most recently received radar image until connectivity is restored. The page polls for WAN restoration every 30 seconds and resumes loading new tiles automatically when internet returns.

26 High Frequency (HF) Propagation Tool http://192.168.50.1/propagation.html

The HF Propagation tool retrieves current band conditions from HamQSL (N0NBH, hamqsl.com) and displays a band-by-band propagation summary. It requires internet access to fetch current data; when offline it shows a model-based estimate rather than live solar data.

BAND	TYPICAL USE IN Emergency Communications (EMCOMM)
160m / 80m	Regional night-time nets, 0–500 mile coverage
40m	Primary EMCOMM HF band — day/night, 500–2000 miles
20m	Long-haul — 2000+ miles, daytime primary
15m / 10m	Long-haul when solar conditions are favorable
6m / 2m	Very High Frequency (VHF) — not covered by HF propagation tool

The tool shows Solar Flux Index (SFI), A-index, K-index, and predicted propagation quality per band. The K-index is the most immediately useful: K 0–2 is excellent; K 3–4 is fair; K 5+ indicates a geomagnetic disturbance that degrades HF propagation.

27 Winlink Radio Email http://192.168.50.1/winlink-import.html

FieldCommand IMS integrates with Winlink — the global radio email network used for Incident Command System (ICS) messaging over RF when internet is unavailable. The integration covers: outbound ICS-213 messages via Winlink, inbound Winlink form import, and Pat (a cross-platform Winlink client) running alongside FieldCommand.

27.1 Winlink Configuration Options

CLIENT	PLATFORM	TRANSPORT
Winlink Express	Windows laptop	VARA High Frequency (HF), VARA FM, Telnet, Pactor
Pat	Raspberry Pi / Linux	VARA FM (via Wine), Telnet, AX.25
RMS Express	Windows laptop	Pactor (requires SCS modem)

27.2 Winlink Form Import

When Winlink messages arrive containing ICS forms (ICS-213, ICS-214, position reports), they can be imported directly into FieldCommand IMS using the Winlink Form Import page.

1. In Winlink Express or Pat, read the incoming message with the ICS form attachment.
2. Save the form as an XML or HTML file.
3. Navigate to **Winlink Form Import** from the dashboard.
4. Click **Choose File** and select the exported form file.
5. Click **Import**. FieldCommand IMS parses the form data and creates the appropriate ICS form entry in the active incident.

28 Amateur Packet Radio Network (AMPRNet) / 44Net Gateway http://192.168.50.1/amprgate.html

■ **This chapter applies only to deployments that include an amateur radio group as the lead or a key partner.** If your deployment is operated entirely by a public safety agency, municipality, or served organization without a licensed amateur radio group involved, the AMPRNet gateway does not apply to you — skip this chapter. FieldCommand IMS runs fully and completely without AMPRNet. All 49 tools, ICS forms, net loggers, FEMA documentation, personnel accountability, and every other feature operate on EMCOMM-NET with no dependency on AMPRNet whatsoever.

AMPRNet — Amateur Packet Radio Network — is the global amateur radio IP network operating on the 44.0.0.0/8 address block permanently assigned by IANA to ARRL for amateur radio use. The ARDC gateway (**amprgw.ampr.org:51820**) provides a WireGuard-encrypted tunnel into that network over the internet. When configured on the dedicated gateway Raspberry Pi (192.168.50.2), every device on EMCOMM-NET gains access to AMPRNet resources — Winlink gateways, APRS-IS servers, and other amateur stations worldwide reachable on 44.x.x.x addresses.

The gateway Pi is completely isolated from the primary FieldCommand server. If it goes down or is not deployed, nothing else on EMCOMM-NET is affected.

28.1 Who Should Deploy AMPRNet?

SITUATION	DEPLOY AMPRNET?
ARES/RACES group leads or co-leads the deployment	✓ Yes — Winlink via AMPRNet path, APRS-IS, inter-node data. Licensed operators handle the registration under their callsign.
Amateur radio club operates the EOC comms section	✓ Yes — same benefits. Club callsign (e.g. your club callsign) is preferable for organizational deployments.
Licensed amateurs are supporting partners with the served agency	■ Consider — only if the licensed operators take full ownership of the AMPRNet registration and ongoing maintenance.
Public safety agency only — no amateur radio group involved	✗ Not applicable — FCC amateur license required for registration. AMPRNet is a Part 97 resource; it cannot be registered or operated by a non-licensed agency.

28.2 The Amateur Radio Group Must Lead This

AMPRNet IP addresses are assigned to a licensed amateur callsign and governed by Part 97. The registration process, ongoing maintenance, and all operational use must be led by the amateur radio group — not the served agency, not the EOC IT department, and not the municipality. The practical steps are:

STEP	WHO
Designate a licensed technical lead (any FCC license class)	ARES EC or club President
Contact the regional AMPRNet coordinator before registering	Technical lead — Illinois: via ARRL Illinois Section
Register at portal.ampr.org using the club or personal callsign	Technical lead

STEP	WHO
Request a /29 subnet (6 usable IPs) or /28 (14 IPs)	Technical lead — allow 2–6 weeks for approval
Download the WireGuard config and configure the gateway Pi	Technical lead — see Installation Guide Step 11
Maintain the portal account and license going forward	Technical lead + designated backup

28.3 What AMPRNet Enables

CAPABILITY	DESCRIPTION
Winlink via AMPRNet path	Reach Winlink RMS gateways at 44.x.x.x addresses without using the commercial internet — keeps message handling within the amateur radio network.
APRS-IS via AMPRNet	APRS-IS servers are reachable on 44.x.x.x. Configure Graywolf or YAAC to use the AMPRNet path instead of the public internet.
Inter-node FieldCommand	Two FieldCommand deployments with 44Net gateways can share net log data and resource status over AMPRNet — no commercial internet required.
Global amateur station reach	Any amateur station worldwide with a 44.x.x.x address is directly reachable from any EMCOMM-NET device.
Permanent static IPs	Your 44.x.x.x block is yours permanently — fixed addresses that never change regardless of ISP or location.

Part 97 applies to all AMPRNet traffic: no encryption of content (WireGuard tunnel encryption of the transport layer is permitted), no commercial traffic, station identification required. All use must comply with Part 97 rules.

28.4 Access Control on EMCOMM-NET

The AMPRNet gateway enforces Part 97 access control: only licensed amateur operators with a valid FCC callsign may authenticate. Callsigns are validated against the offline FCC database. All access attempts are logged to /var/log/amprgate-access.log with callsign, IP, timestamp, and result. Session tokens have an 8-hour TTL.

28.5 Gateway Status Page

The AMPRNet status page (amprgate.html) shows tunnel up/down status, your assigned 44.x.x.x address, connected peers, and the access log. The public read-only port (9000) is accessible from any EMCOMM-NET device. Tunnel control is restricted to the gateway Pi console (port 9001, localhost only) — an operator must be physically at the gateway Pi keyboard to bring the tunnel up or down.

Full AMPRNet setup procedures — WireGuard configuration, IP forwarding, route advertisement, and verification checklist — are in Installation Guide Step 11. Portal registration: portal.ampr.org. WireGuard gateway endpoint: amprgw.ampr.org:51820.

29 JS8Call — High Frequency (HF) Digital Messaging

http://192.168.50.1/

JS8Call is a weak-signal HF digital mode designed for keyboard-to-keyboard messaging and store-and-forward relay. It operates on a Windows laptop connected to the incident's HF transceiver (e.g. an IC-7300 with a Digirig or similar Universal Serial Bus (USB) audio interface) alongside the FieldCommand system on the same EMCOMM-NET network.

29.1 JS8Call for Emergency Communications (EMCOMM)

CAPABILITY	DESCRIPTION
Keyboard messaging	Real-time chat-style messaging between stations — no callsign required to copy
Store and forward	Messages are stored at relay stations and forwarded when the destination is heard
Heartbeat beacons	Automatic periodic beacons announce your station and provide propagation data
Group messaging	Messages addressed to @EMCOMM, @ARES, @RACES reach all monitoring stations
SNR as low as -24dB	Copies signals far weaker than SSB voice or Winlink VARA

29.2 Integration with FieldCommand

JS8Call traffic is monitored on the EMCOMM-NET network. Incoming messages that match Incident Command System (ICS) message formats can be manually entered into the ICS-213 or the Net Control Logger. A future integration (planned for a subsequent release) will allow direct JS8Call message ingestion into the incident log.

30 National Traffic System (NTS) Radiogram Generator

http://192.168.50.1/nts.html

The NTS Radiogram Generator produces properly formatted ARRL National Traffic System radiograms for formal message handling. NTS radiograms are the standard for health-and-welfare and priority traffic during disasters when telephone and internet are unavailable.

30.1 Radiogram Fields

FIELD	DESCRIPTION
Precedence	ROUTINE · WELFARE · PRIORITY · EMERGENCY
Handling	HXG (delivery by mail), HXC (report delivery), others
Station of origin	Originating callsign
Check	Word count of the text — calculated automatically
Place of origin	City/state of the originating station
Time/Date	When the message was filed — filled automatically
To / Phone	Addressee name and phone number for final delivery
Text	Message body — 25 words maximum for efficient traffic handling
Sent by	Callsign of the operator sending this copy

Click **Generate Radiogram** to produce a formatted radiogram ready to transmit. The word count (check) is calculated automatically. Click **Print** to print on a standard traffic form.

31 Repeater Database http://192.168.50.1/repeaters.html

The Repeater Database stores VHF/UHF repeater information for your deployment area. It feeds the ICS-205 communications plan channel picker, the channel library, and the cheat sheets. Data is stored locally — no internet required once loaded.

31.1 Loading Repeater Data

The page has three source tabs: Offline File (recommended), Server API, and Demo Data. The Offline File tab is the practical method for most deployments.

METHOD	HOW TO USE	REQUIREMENTS
Offline File (recommended)	Log in at repeaterbook.com → search your area → Export → CSV. Drag the downloaded CSV onto the drop zone, or click to browse.	Free repeaterbook.com account. No API token needed — any logged-in user can export a CSV.
RepeaterBook app	Search your area in the RepeaterBook mobile app → Export → Share file. Transfer the file to a device on EMCOMM-NET and load it.	Free RepeaterBook app (iOS or Android).
Server API (optional)	Run fetch_repeaters.py on the FieldCommand Pi. Place your token in /opt/fieldcommand/data/repeaterbook_token.txt then run: python3 scripts/fetch_repeaters.py	Approved RepeaterBook API token — this is a separate application process from a regular account login. Apply at repeaterbook.com/api/token_request.php. Allow several weeks.
Manual entry	Click + Add Repeater and fill in fields. Best for a small number of local repeaters.	None

■ The RepeaterBook API token is not the same as a RepeaterBook account login. It requires a separate application and approval by RepeaterBook staff. The CSV export method works with any free account and is the recommended approach for field deployments. The CSV can be refreshed before each activation by any operator with a repeaterbook.com account.

31.2 Repeater Fields

FIELD	DESCRIPTION
Callsign	Repeater trustee callsign
Output frequency	Receive frequency (what you listen on)
Offset	± MHz offset from output, or simplex
CTCSS / DCS	Tone or digital code required to access the repeater
Mode	FM · D-STAR · Fusion (C4FM) · P25 · DMR
Coverage	Coverage description — city, county, or regional
Notes	Access restrictions, linking, EchoLink node, etc.

FIELD	DESCRIPTION
Emergency use	Flag as preferred for emergency use — highlighted in ICS-205

1. Click **+ Add Repeater**.
2. Fill in applicable fields. Callsign, output frequency, and offset are required.
3. Click **Save**. The repeater is immediately available in the ICS-205 channel picker and the cheat sheets.

31.3 Map View — Plotting Repeaters on a Map

Once repeater data is loaded, the ■ **Map View** button in the toolbar opens a full-screen Leaflet map showing every repeater in the current filtered view that has GPS coordinates. RepeaterBook CSV exports include latitude and longitude for every repeater — these are stored in the database and used for map plotting automatically.

MAP FEATURE	DESCRIPTION
Pin colors by mode	Green = FM · Blue = D-STAR · Orange = C4FM/Fusion · Purple = DMR · Red = P25 · Amber = ARES/RACES-affiliated (overrides mode color)
Filter sync	The map plots only the repeaters currently visible in the table — applying a band, mode, state, or ARES filter also filters the map pins. Set filters first, then open the map to see exactly the repeaters you want.
Pin popup	Click any pin to see: frequency, callsign, offset, CTCSS/DCS tone, mode, location, county, sponsor, notes, and ARES/RACES/SKYWARN badges.
Add to Channel Library	The popup includes a "+ Add to Channel Library" button — click it to immediately import that repeater as a channel.
Count display	The Map View button shows the count of plottable repeaters in parentheses — e.g. "■ Map View (247)". Repeaters without coordinates are not plotted but are still shown in the table.

1. Load repeater data using any method (CSV, app, or manual entry).
2. Optionally apply filters — band, mode, state, ARES/RACES — to narrow the view.
3. Click ■ **Map View (N)** in the toolbar. The map opens centered and zoomed to fit all visible repeaters.
4. Click any pin to see the full repeater details and import it to the channel library.
5. Press **Escape** or click ✕ **Close** to return to the table.

31.4 Repeater Overlays on Tactical and Resource Maps

The Tactical Map (tactical.html) and Resource Map (resource_map.html) both include a ■ **Repeaters** layer toggle. This overlay fetches live repeater data from the server and plots pins on top of the APRS traffic, resource positions, or field unit locations — giving situational awareness of which repeaters are available near a specific area or resource.

MAP	HOW TO ENABLE	USE CASE
Tactical Map (tactical.html)	Click ■ **Repeaters** in the layer toggle bar. First click fetches and plots all repeaters. Subsequent clicks toggle the layer on/off without re-fetching.	See which repeaters cover the areas where APRS stations are active. Identify the closest ARES/RACES-linked repeater to a field unit.

MAP	HOW TO ENABLE	USE CASE
Resource Map (resource_map.html)	Click ■ **Repeaters** in the toolbar. Toggle on/off at any time.	Find coverage for deployed resources. Verify repeater access from a staging area or shelter location.

> ■ The repeater overlay loads from the FieldCommand server database — it reflects exactly the repeater data imported on the Repeater Database page. If the overlay shows no pins, import repeater data at repeaters.html first. Only repeaters with GPS coordinates are plotted; repeaters entered without coordinates appear in the Repeater Database table but not on the map.

32 Channel Library http://192.168.50.1/channel_library.html

The Channel Library stores pre-configured communications channels for common operations: mutual aid interoperability, public safety talk groups, amateur simplex and repeater frequencies, and satellite channels. Channels from the library can be inserted directly into the ICS-205 Communications Plan with a single click.

Each channel entry stores: channel name, frequency/talk group, mode, CTCSS tone, purpose, and Division / Group assignment. The library is pre-populated with common interoperability channels and can be customized for your jurisdiction's channel plan.

> NIFOG (National Interoperability Field Operations Guide) channels and VTAC channels are included in the default library. Your jurisdiction's specific mutual aid channels should be added during initial system configuration.

33 Hospital Proximity & Facilities Directory http://192.168.50.1/hospitals.html

The Hospital Proximity page and Facilities Directory store the locations and contact information for medical facilities, EOCs, staging areas, and other operational facilities used during incidents.

33.1 Hospital Directory

FIELD	DESCRIPTION
Facility name	Official hospital or clinic name
Address	Street address for navigation
Trauma level	Level I · II · III · IV · not designated
Phone	Main switchboard and emergency department numbers
Helipad	Yes / No — for aeromedical considerations
Distance	Calculated from your station grid square (approximate)
Notes	Diversion status, special capabilities, current capacity

33.2 Facilities Directory

The Facilities Directory (facilities.html) stores staging areas, EOCs, shelters, supply depots, and other operational locations. Facilities appear in the ICS-205 and can be linked to T-card assignments. Each facility stores name, address, type, capacity, and operational notes.

34 Reference Tools — Grid, Cheat Sheets, Resources, Print Center

FieldCommand IMS includes a set of reference and utility tools used throughout incident operations. All operate fully offline.

34.1 Grid Square Calculator

The Grid Square Calculator (grid.html) converts between Maidenhead grid locators and decimal coordinates, and calculates distance and bearing between two grid squares. It supports 4-character (e.g. EN52) and 6-character (e.g. EN52ab) precision.

FUNCTION	INPUT	OUTPUT
Grid to coordinates	Maidenhead grid square (4 or 6 char)	Decimal lat/lon and DMS
Coordinates to grid	Decimal lat/lon	Maidenhead grid square
Distance and bearing	Two grid squares	Distance (km/mi) and true bearing

34.2 Radio Cheat Sheets

The Cheat Sheets page (cheatsheets.html) provides quick-reference cards for common amateur radio and Incident Command System (ICS) procedures: phonetic alphabet, Q-codes, RST signal report scale, ICS command structure, and common High Frequency (HF) calling frequencies. All cards are printable and designed to fit on a single laminated sheet. (National Incident Management System (NIMS) resource typing is not a cheat-sheet card — it has its own dedicated page, resource_types.html.)

34.3 NIMS Resource Typing Library

The NIMS Resource Typing Library (resource_types.html) contains the standard NIMS resource type definitions used to populate T-cards. Custom resource types specific to your organization can be added here and immediately become available on the T-card board.

34.4 ICS Position Checklists

The Position Checklists page (position_checklists.html) provides NIMS-standard activation checklists for each ICS position: Incident Commander (IC), Safety Officer, Operations, Planning, Logistics, Finance, and key unit leaders. Each checklist can be checked off during position activation and printed for the position binder.

34.5 Print Center

The Print Center (printcenter.html) provides optimized print layouts for all ICS forms and documents. From any device on EMCOMM-NET, select the document and click Print — printing uses your device's own browser print dialog (you can print to any printer that device can reach, or save as PDF). There is no server-side (CUPS/remote) print spooling.

35 Network Hardware — Routers, Switch, and Coverage Extension

FieldCommand IMS is designed to work with standard Wi-Fi routers, managed Power over Ethernet (PoE) switches, and mesh networking to provide reliable wireless coverage across a deployment site. This chapter covers the recommended hardware configuration and how to extend coverage using AiMesh nodes.

35.1 Recommended Router — ASUS RT-BE58 Go

The ASUS RT-BE58 Go is the recommended primary router for FieldCommand IMS. It is a compact, portable Wi-Fi 7 router designed for mobile deployments — it runs on USB-C power, supports AiMesh for seamless coverage extension, and provides dual Wide Area Network (WAN) ports for simultaneous cellular and satellite sources. It is available from most electronics retailers for approximately $100–130.

SPEC	RT-BE58 Go
Wi-Fi standard	Wi-Fi 7 (802.11be) — backward compatible with all devices
Bands	2.4 GHz + 5 GHz dual-band
WAN ports	2× — one for cellular/primary, one for satellite/secondary
Power	USB-C — runs from any 65W USB-C PD adapter or battery bank
AiMesh support	✓ — extends EMCOMM-NET coverage with additional RT-BE58 Go units
Typical range	~2,500 sq ft indoors / larger outdoors

35.2 Extending Coverage with AiMesh

AiMesh is ASUS's mesh networking system. Additional ASUS RT-BE58 Go units can be added as AiMesh nodes to extend the EMCOMM-NET signal to additional rooms, floors, or outdoor areas. All nodes use the same Service Set Identifier (SSID) and password — devices roam between nodes automatically without reconnecting. Nodes are connected to the primary router via wired Ethernet through the UniFi switch for best performance.

1. Connect the AiMesh node to the UniFi switch with an Ethernet cable.
2. Power on the node. It will appear in the ASUS Router app or the primary router's admin interface at 192.168.50.254.
3. In the router admin interface, navigate to **AiMesh** and click **Add AiMesh Node**. The new node appears automatically.
4. Click **Connect**. The node joins the mesh and begins broadcasting EMCOMM-NET within about 60 seconds.
5. Position nodes to overlap coverage by 20–30% for seamless roaming.

35.3 UniFi Switch Lite 16 PoE

The UniFi Switch Lite 16 PoE is the recommended wired backbone switch. It provides 16 ports with PoE+ on 8 ports — powering PoE devices (cellular antennas, cameras, PoE access points) without separate power adapters. The UniFi management interface provides per-port visibility of link status, speed, and PoE power draw, which is useful for diagnosing connectivity issues in the field.

PORT GROUP	RECOMMENDED USE
Ports 1–2	FieldCommand Pi 5 (primary server) and 44Net Gateway Pi 5
Ports 3–4	ASUS RT-BE58 Go primary router LAN and AiMesh node uplinks
Ports 5–8	PoE — cellular antenna, satellite PoE injector, PoE cameras
Ports 9–16	Operator workstations (Pi 500 units, Windows laptop, tablets)

35.4 Power Considerations

For field deployments without shore power, FieldCommand IMS can run from two Astron RS-35M-AP regulated linear power supplies (one per Pi cluster) fed from a portable generator, or from a high-capacity LiFePO4 battery with a pure-sine inverter. The complete system draws approximately 80–120 watts under typical load. For battery-only operation, a 100Ah 12V LiFePO4 battery provides approximately 8–10 hours of runtime.

36 Appendix — Quick Reference & Administration

A.1 All Pages — Uniform Resource Locator (URL) Reference

URL	DESCRIPTION
192.168.50.1/index.html	Main Dashboard
192.168.50.1/incident.html	Incident Management / Command Section
192.168.50.1/incident_mgmt.html	Incident Archive, Restore, Delete, Beta Reset
192.168.50.1/event_templates.html	Pre-Planned Event Templates
192.168.50.1/resources.html	Resource Board (flat list)
192.168.50.1/ics/operations.html	T-Card Resource Board (drag-and-drop)
192.168.50.1/resource_map.html	GPS-Tracked Resource Map
192.168.50.1/resource_types.html	National Incident Management System (NIMS) Resource Typing Library
192.168.50.1/checkin.html	Manual Check-In (ICS-211)
192.168.50.1/scan_checkin.html	Quick Response (QR) code / Barcode Scan Check-In
192.168.50.1/roster.html	Member Roster and QR Code Generator
192.168.50.1/netcontrol.html	Amateur Radio Net Control Logger
192.168.50.1/starcom.html	Public Safety Net Logger
192.168.50.1/observer.html	Observer Mode — Read-Only Net View
192.168.50.1/deadmans.html	Dead Man's Switch
192.168.50.1/iap.html	Incident Action Plan (IAP) Assembly — Planning Section
192.168.50.1/iap_compile.html	IAP One-Click Portable Document Format (PDF) Compilation
192.168.50.1/ics-form.html	Incident Command System (ICS) Form Suite — all forms
192.168.50.1/ics213.html	ICS-213 General Message
192.168.50.1/ics214.html	ICS-214 Activity Log
192.168.50.1/ics309.html	ICS-309 Communications Log
192.168.50.1/fema_costs.html	Federal Emergency Management Agency (FEMA) Public Assistance (PA) Cost Documentation
192.168.50.1/fema_rates.html	FEMA Equipment Rate Schedule
192.168.50.1/cost_dashboard.html	Real-Time Cost Dashboard
192.168.50.1/wan_settings.html	Wide Area Network (WAN) Source Configuration

URL	DESCRIPTION
192.168.50.1/wan-status.html	WAN Status Detail Page
192.168.50.1/radar.html	Animated Next Generation Radar (NEXRAD) Radar
192.168.50.1/propagation.html	High Frequency (HF) Propagation Data
192.168.50.1/tactical.html	Tactical Automatic Packet Reporting System (APRS) Map
192.168.50.1/resmap.html	Public Safety Resource Map
192.168.50.1/callsign.html	Federal Communications Commission (FCC) Callsign Lookup
192.168.50.1/amprgate.html	Amateur Packet Radio Network (AMPRNet) (44Net) Gateway Status
192.168.50.1/nts.html	National Traffic System (NTS) Radiogram Generator
192.168.50.1/winlink-import.html	Winlink Form Import
192.168.50.1/briefing_204a.html	ICS-204A Briefing Sheet
192.168.50.1/hospitals.html	Hospital Proximity Directory
192.168.50.1/facilities.html	Facilities Directory
192.168.50.1/repeaters.html	Repeater Database
192.168.50.1/channel_library.html	Channel Library
192.168.50.1/cheatsheets.html	Radio / ICS Cheat Sheets
192.168.50.1/grid.html	Grid Square Calculator
192.168.50.1/position_checklists.html	ICS Position Checklists
192.168.50.1/meetings.html	Meeting Scheduler
192.168.50.1/printcenter.html	Print Center
192.168.50.1/sartopo_import.html	SARTopo GeoJSON Import
192.168.50.1/preflight.html	Preflight Deployment Checklist
192.168.50.1/refs.html	Reference Library
192.168.50.1/setup.html	Organization Setup
192.168.50.1/general_info.html	General Info / ICS-201

A.2 API Ports

PORT	SERVICE	FUNCTION
5050	fcc_lookup_server.py	FCC lookups · nets/roster · hospitals · repeaters · resource types · facilities · GPS/dead-man switch · WAN config
5051	health_monitor.py	System health — CPU/memory/disk, service states, connectivity roll-up

PORT	SERVICE	FUNCTION
5055	ics_platform_server.py	ICS forms · incidents · T-cards · check-ins · IAP · FEMA costs
5056	reference_server.py	Offline reference library (renders PDFs)
8083	tile_server.py	Offline map tiles (MBTiles)
9000	amprgate_status.py	44Net status — read-only on EMCOMM-NET (gateway Pi, 192.168.50.2)
9001	amprgate_status.py	Tunnel control — localhost only (gateway Pi)
80	nginx	Serves all HTML pages and static assets

A.3 Background Services (systemd)

SERVICE FILE	FUNCTION
ics-platform.service	ICS platform server (port 5055)
fcc-lookup.service	FCC lookup and config server (port 5050)
health-monitor.service	System health monitor (port 5051)
fieldcommand-refs.service	Offline reference library server (port 5056)
fieldcommand-tiles.service	Offline map tile server (port 8083)
deadmans.service	Per-net dead-man switch monitor
wan-monitor.service	WAN source monitoring and failover
aprs-rf.service	RF APRS receive via direwolf or Terminal Node Controller (TNC)
amprgate-poll.service	44Net tunnel keepalive and reconnect (gateway Pi)
kiwix.service	Kiwix offline reference library server
backup.service / backup.timer	Nightly SQLite backup to Universal Serial Bus (USB) drive

A.4 Copyright & License

A.5 Abbreviations & Acronyms

The following abbreviations and acronyms are used throughout this manual. Each is defined on first use in every chapter. This table serves as a quick reference for readers who open to a chapter mid-document.

AMPRNet	Amateur Packet Radio Network (44.0.0.0/8)	**NWCG**	National Wildfire Coordinating Group
AP	Access Point (Wi-Fi)	**NWS**	National Weather Service
API	Application Programming Interface	**OSC**	Operations Section Chief
APRS	Automatic Packet Reporting System	**P25**	Project 25 (APCO) - digital public safety radio standard
ARES	Amateur Radio Emergency Service	**PA**	Public Assistance (FEMA grant program)
ARRL	American Radio Relay League	**PAR**	Personnel Accountability Report
AUXCOMM	Auxiliary Communications	**PDF**	Portable Document Format
CGNAT	Carrier-Grade Network Address Translation	**PIO**	Public Information Officer

COML	Communications Unit Leader (Logistics Section)	**PO**	Purchase Order
CTCSS	Continuous Tone-Coded Squelch System	**PSC**	Planning Section Chief
CUPS	Common Unix Printing System	**PW**	Project Worksheet (FEMA PA documentation form)
DCS	Digital-Coded Squelch	**PoE**	Power over Ethernet
DHCP	Dynamic Host Configuration Protocol	**QR**	Quick Response (two-dimensional barcode format)
DIVS	Division/Group Supervisor	**RACES**	Radio Amateur Civil Emergency Service
DMOB	Demobilization Unit Leader (Planning Section)	**RAID**	Redundant Array of Independent Disks
DMR	Digital Mobile Radio	**RESL**	Resources Unit Leader (Planning Section)
EMCOMM	Emergency Communications	**RRCC**	Regional Response Coordination Center
EOC	Emergency Operations Center	**SAR**	Search and Rescue
ETA	Estimated Time of Arrival	**SFI**	Solar Flux Index
FCC	Federal Communications Commission	**SITL**	Situation Unit Leader (Planning Section)
FEMA	Federal Emergency Management Agency	**SOFR**	Safety Officer (ICS Command Staff)
FSC	Finance/Administration Section Chief	**SSD**	Solid-State Drive
GPS	Global Positioning System	**SSID**	Service Set Identifier (Wi-Fi network name)
GSUL	Ground Support Unit Leader (Logistics Section)	**STLD**	Strike Team Leader
HF	High Frequency (3-30 MHz radio)	**SWPC**	Space Weather Prediction Center (NOAA)
IAP	Incident Action Plan	**TFLD**	Task Force Leader
IC	Incident Commander	**TNC**	Terminal Node Controller (packet radio hardware)
ICS	Incident Command System	**TTL**	Time To Live (session expiration interval)
JSON	JavaScript Object Notation	**UC**	Unified Command
LAN	Local Area Network	**UHF**	Ultra High Frequency (300 MHz-3 GHz radio)
LOFR	Liaison Officer	**ULS**	Universal Licensing System (FCC amateur database)
LSC	Logistics Section Chief	**URL**	Uniform Resource Locator (web address)
MAC	Multi-Agency Coordination	**USB**	Universal Serial Bus
MEDL	Medical Unit Leader (Logistics Section)	**USCG**	United States Coast Guard
NEXRAD	Next Generation Radar (NWS WSR-88D network)	**VARA**	VARA (Winlink HF/FM digital mode modem software)
NIFOG	National Interoperability Field Operations Guide	**VHF**	Very High Frequency (30-300 MHz radio)
NIMS	National Incident Management System	**VTAC**	VHF Tactical Interoperability Channel
NRCC	National Response Coordination Center	**WAN**	Wide Area Network (internet connection)
NTS	National Traffic System (ARRL)	**WireGuard**	WireGuard (modern VPN tunneling protocol)
NVMe	Non-Volatile Memory Express (fast storage interface)		